Ojo Alamo Sandstone | Research Source Brief

Peace Engineering • Cuba, New Mexico • Source assessment

Source and relevance

USGS, "Hydrogeology of the Ojo Alamo Sandstone in the San Juan Structural Basin, New Mexico, Colorado, Arizona, and Utah" (1990). The uploaded publication is a one-page map plate containing regional hydrogeologic maps and explanatory text. It is relevant to Cuba as regional geological context, but does not establish that Cuba's operating municipal wells draw from the Ojo Alamo Sandstone.

Research questions supported

Question	How to use the publication
Q4 — Compare Ojo Alamo and San Jose	Strong for the Ojo Alamo side: maps of sandstone extent/structure, depth to its top, thickness, and mapped wells. A separate San Jose Formation source is needed for comparison.
Q2 — Well depth, water depth, aquifer units	Partial: helps identify Ojo Alamo geometry and mapped wells, but does not provide a comprehensive table of Cuba municipal well depths or water-level measurements.
Q5 — Groundwater quality / salinity	Limited: any regional discussion is contextual; the plate is not a Cuba-specific water-quality time series.
Q3 — Is Cuba approaching depletion?	Not supported directly: no confirmed Cuba municipal pumping history, groundwater-level trend, recharge budget, or usable-storage estimate.

Recommended figures for the report

Use the publication's regional maps showing (1) depth to the top of the Ojo Alamo Sandstone, (2) sandstone thickness/configuration, and (3) wells completed in the formation. Mark Cuba's approximate location only after checking the map scale and coordinate reference. Label these as historical regional interpretations, not current groundwater conditions.

Guidance for the classmate preparing plots

This publication is primarily a mapped geological reference—not a ready-to-plot CSV. A useful deliverable is an annotated map or side-by-side geological comparison with San Jose Formation information. Do not infer groundwater depletion, water quality, or Cuba's municipal source aquifer from the maps alone. Confirm municipal well completion formations through Village of Cuba or NM Office of the State Engineer well records.

Source: https://www.usgs.gov/publications/hydrogeology-ojo-alamo-sandstone-san-juan-structural-basin-new-mexico-colorado-arizona